ActInf Textbook Group ~ Cohort 3 ~ Meeting 5 (Chapter 2, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 5 (Chapter 2, part 2) | 58:05

Hello, it's February 22nd, 2023. We're in meeting five for cohort three. We're in our second
discussion on chapter two. Today we'll be mostly looking at questions from chapter two,
hearing people's thoughts on it, written or whatever they want to bring up today.
And then in the last few minutes, we'll look ahead to chapter three. First though,
wanted to note underneath live meetings on February 27th in five days, we will have Thomas
Parr join for a live stream, hashtag first author. So it's at 17 UTC for likely an hour. If anybody
wants to join the live stream itself, then just email active inference at Gmail and let us know.
And we'll add you to the calendar event. If you want to watch a long live, here's the, or share it.
And this is like the watch and the rewatch link. And then if you have any specific questions,
it could be something that we've already raised, or it could be something more general slash personal.
So just before we jump to chapter two, what does anyone think would be interesting or meaningful to ask Thomas?
Feel free to think about it or type it here. But basically when the day comes, it's going to be whoever
has, thank you, Jonathan. It's going to be whoever's there in the moment. So just email if you want to
join on that stream and whichever questions are here slash questions that we've just curated.
And then if there's time or if anyone is watching live, they can put a question in the live chat too.
So we'll continue to explore like what kinds of engagements with Thomas, Giovanni, and Carl
are best for the book and everything, but this is going to be a lot of fun.
Okay.
Well, we have many chapter two questions. Does anyone have a specific one they want to
jump into first or just any chapter two reflection or comment that they want to share?
There's totally time to do that if you want to.
Okay.
Okay.
Let's just try to pick off some comments and questions.
Kind of just drawing, drawing from the question posterior.
How does active inference go beyond the recognition that action and perception have the same inferential
nature?
Page 24.
What do people think?
So this is probably referring to section 2.4.
How do we go beyond recognizing, well, maybe perception and learning have a similar inferential nature?
Maybe we can have a unified framework that models perception and action together.
And maybe they even have a unified imperative.
I think the simplest answer is they're different parameters.

That's how we move past that recognition.

They're modeled as different parameters in a unified model.

So we want to add another thought or question on that.

Okay.

Okay, anyone else want to pick a specific question to go to?

Otherwise, we have many that we can come to.

Okay.

Why express potential energy and expected free energy in the space of log probabilities on page 33?

Consistent with the notion of potential energy in physics, expected free energy is expressed in the space of log probabilities.
Nice question.
Any first thoughts?
Let's look at where it was in the textbook.
Okay.
So we're talking about taking the expected free energy of each policy.
So equation 2.6, expected free energy over policies.
That's sharpening our action prior, our habits, into the action posterior that we're actually going to sample our actions from.
Does anyone want to give a thought on that question?
Yeah.
So I'm in one of my math modules.
We're currently studying Bayesian maths.
So one of the benefits of using log probabilities is it makes it easier to calculate derivatives when you're modeling Gaussians.
The second point was computational benefit.
You might encounter floating point errors when dealing with, I think, marginal evidence, which can be very small for some values.
So calculating the log version of those reduces the likelihood of getting into encountering floating point issues and precision errors.
That's what I'm trying to say.
Awesome.
Thank you.
Yeah.
The log is something that's like very simple, whether it's like a log base 2 or LN, the natural log.
It, it does a lot.
And it also has one foot in the kind of computational tractability benefits, like being able to keep track of small probabilities.
Like if you have a hundred options, then most of them are going to be small, like less than 1%.
Now, if you have a thousand options, most of them are going to be small, like on average less than a 10th of a percent.
So as you get many options or probability distributions, you want to be able to keep track with high resolution of these very small numbers on real computers.
So it helps purely pragmatically in the kinds of computers we have today.
Then a little bit more mathematically, the log connects to Jensen's inequality and also to

information theory and self-surprisal.
And then something that's unpacked a lot more in Bayesian physics and less in this textbook itself is like physics uses kind of like gravity is like a self-information.
Not exactly, you know, read where it's actually connected, but it turns out that like the ball rolling to the bottom of the hill to minimize the potential energy has a lot to do with the quote ball rolling to the bottom of the hill of a statistical distribution.
If we think about self-surprise, self-information in a log probability space.
Ali?
Ali?
Yes.
And one more advantage of using log probabilities is to, or more generally to use logs instead of the actual parameters, is to convert some of the multiplications to addition and hence trying to convert at least parts of the nonlinearities of the equations to linear equations.
So in that case, well, of course, we know dealing with linear equations is much more tractable and also much more, I mean, linear techniques has much more developed than just dealing with directly with nonlinear equations and nonlinear calculations.
Thanks.
Yeah.
Two of the log properties that come into play a lot and you can go back and forth, like they're used in both directions.
Here, log of m times m, so log of 5 times 10, log of 50, equals log of 5 plus log of 10.
So you can break multiplication inside of the parentheses, you can break it into two logs with the same a value.
And analogously, if you have like log m minus log n, it's the same as log of m divided by n.
And these kinds of maneuvers and rewritings are used greatly.
And it turns out that the KL divergence, which is one divergence measure that's used.
So just like you'll know that you're doing better in the least squares, sum of squares, linear regression world.
You know that you're doing better when the sum of squares is going down.
In the KL divergence world, variational inference world, your model is getting better fit when the KL divergence is going down.
Again, there's more detail, but these all are empowered by the ability to have something within a log that's multiplied, which also gives this the interpretation as like potentially a self-surprisal self-information and then separate that out and do addition.
So it has computational benefits.
It has statistical benefits and has also a deeper connection to Bayesian physics.
Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Sorry.

Before going into the next question, I just wanted to comment on the previous question,

the question number two.

Sorry, I raised my hand a little bit later.

So you didn't notice, but yeah, about the relation between the perception and action.

So very briefly, Alan Berthos actually proposes that perception is simulated action.

So it basically implies that perception and motion are kind of irreducibly interconnected.

And actually, he also claims that the former, I mean, the perception defines the action.

So, yeah, it's also another viewpoint, which is way before the development of active inference and pre-energy principle.

But these ideas of the interconnection between the action and perception has been discussed all, I mean,

throughout the history of cognitive science and especially the last few decades in terms of embodied cognition and so on.

But in my view, Alan Berthos' claim about this specific interconnection between perception and action and basically making them equivalent to each other in terms of their,

one of which being the simulation of the other is an interesting viewpoint to have in mind beyond the inferential nature of them.

Nice. Yeah.

Like even qualitatively, we can draw connections with different threads.

Like Helmholtz, perception is unconscious inference.

Okay.

No.

Just save that.

Thank you for that.

Okay.

Now, what's the pragmatic turn?

Well, that's about in the Bayesian brain.

It's about thinking about action alongside inference.

So it's like, okay, we have Helmholtz, perception is unconscious inference.

Now let's extend that to action and perception as unconscious inference.

That's still a qualitative idea.

And ACDIF and FEP are helping us develop quantitative methods that are compatible with those kind of qualitative ways of thinking about perception, cognition, and action.

Okay.

Upper bound, I think we came to last week.

So.

Okay.

So.

What is meant by bounded computational and mnemonic resources?

Anyone have a thought on this or related idea?

I think this is a great answer.

Just.

You can write an equation, you know, 10 to the 10 to the 10 to the 10 to the 10 to the 10 to the 10 to the 10 power.

But just because you can write it on a piece of paper doesn't mean that there's like a calculator that has that much RAM.

It might just, you know, the number of digits that you need to store might be outside of the heat capacity of Earth or something like that.

So.

We want to be able to express models in an open-ended way in principle.

So.

So.

Also.

Especially if we're talking about what kind of inference is optimal.

It's important to take this probably approximately correct approximate Bayesian computation heuristics angle that respects the actual time space trade-offs that are fundamental for any actual system and any model that we make of a given system.

Like.

It's not behaving accurately.

You know.

If it remembered every single pixel.

It's like, well, yeah, but there's not enough space for it to remember every pixel.

So it might be doing something that's actually very adaptive involving forgetting the vast majority of information.

And there's, there's no trade-off there, but if we can only state the models in principle and we don't keep it bounded with the limitations of real systems.

And the approximate Bayesian computation framework, then, you know, maybe the scripts don't run because.

And that's also really, yeah, there's other angles there.

Any thoughts or comments on this or this is a good answer.

Any thoughts or comments on this or this is a good answer.

Also kind of a nice link to mnemonic resources like memory palace.

Like, what if you didn't have to remember every single word in the poem, but you remembered some nested model where the higher part of the nested model involved touching parts of the environment or walking through an imagined space.

And that sounds like a weird sounds with a

How does active inference incorporate feelings into the framework theory?

In the apple frog example, when the human believes she is confronted with an apple before the observation,

is she feeling hungry or low on glucose?

When she subsequently observes the thing jump, does that observation include the feeling of disappointment,

the release of stress hormones, etc.?

What would anyone say on this?

Jonathan, and then anyone else?

Yeah, so I wrote the thing below there that I think that there's a strong relationship there between feelings and observations,

in the sense that we observe our bodily states.

But I think this ties in nicely with some of Mark Solon's work on affect and consciousness.

Yeah, so I think that, yeah, once you think about bodily states, and the inferences we make about the state that we're in based on those observations.

Nice, thank you.

Anyone else?

I would put this in the counter.

Oh, yeah, Ali first.

Go for it.

I just wanted to point out that there's actually a theory of emotions or feelings developed within active inference framework,

namely deeply felt affect theory developed by Casper Hesp and Ryan Smith and others.

So if anyone's interested in exploring how we can theoretically frame feelings purely in terms of active inference,

I would highly recommend reading this fascinating paper from, I guess, 2020.

Yes, thank you.

I would put it in the category of possible to model.

Like, you could make a model about it, you could write a paper about it,

but not intrinsic within the simplest kernel.

Because feelings also, as Solms will be keen to point out, feelings implies a feeler.

And so that is bringing in the qualia, phenomenology, experience, awareness topic.

But the model can also be stated without those considerations playing an outsized role.

But if someone's making a map, like a generative model, and they want to include those variables,

there are some strategies that people have been implementing for years to engage with affective states in a computational psychiatry approach.

Like, and just an empirical strategy might be like, we start with a symbol model.

And then we find, oh, there's like this interesting residual pattern where people are overestimating how likely apples are around noon.

And then we get a new sensor in our laboratory.

Now we're measuring glucose.

Or we are getting some health data.

So now we can differentiate people who have this metabolic condition from that.

And we're finding that they have a different inference strategy.

So it's not about just sketching it and just saying, oh, it's a done deal because it's on paper.

It's about that iterative modeling and a plurality of possible ways that feelings could be implemented.

Though I think Solms and others will say that they have in silico feeling agents.

So that, again, opens up a whole other level of discussion.

All right.

When quantifying dissimilarity between the prior and posterior, so for example, Bayesian surprise.

So, surprise alone, just surprisal, is about one data point coming in, how surprised are we?

Measured in information theoretic units, like a nat or a bit.

But if we're talking about Bayesian surprise, we're talking about how much the prior shifts to become the posterior as a function of the observation coming in.

Why do we choose the KL divergence over other measures?

Is it just a canonical choice?

Does it matter?

Since KL divergence is asymmetric, is there any reason we have chosen this particular order?

Is there any intuition motivating the choice?

Does anyone want to give a first thought?

Okay.

One example of an alternative measure.

Here's with Sajid et al.

This is getting kind of as far as we might want to visit it here, but it's important to note the KL divergence, as the questioner notes.

It's not actually a measure in the proper metric sense.

And I don't mean like European metric system.

Like, the measurement between A and B has to be the same between B and A for it to be a distance measure.

Whereas the KL divergence, it is a quantitative value, but the divergence from A to B is not necessarily B to A.

In fact, it's generally not.

So KL divergence is an asymmetric divergence quantification.

Now, it's possible to have a very closely related divergence called the Jefferies, which is just the KL plus the other direction.

So it calculates like it's basically the KL of A to B plus the KL of B to A.

And so that is symmetric.

Hence, that is a real distance measure.

Other measures have been proposed.

The Rainey.

And they may discuss other options as well.

Why is it used?

In the context of variational Bayesian inference and probability, it just turns out that the log properties work.

And in one of the earlier equations, which maybe Ali will remember specifically, like, Ali, go for it.

And I'll think about which equation is the one with the KL definition.

It was equation 2.3.

And again, also, there is a lemma called Neyman-Pearson lemma, which actually proves that the most efficient way to distinguish between two probability distribution based on a given observation is KL divergence.

So it's not just an arbitrary choice, but I believe it's a proven theorem.

I mean, it's the most efficient method for comparing those two probability distributions.

Thank you.

Also, just to kind of give one related angle to connect, like, why this whole discussion matters, for those who saw that as in the weeds, it definitely is.

The way that pragmatic value is going to be determined is ultimately through something.

KL is used for information and on the pragmatic side.

But this is how, when we talk about expectation and preference, and we want to evaluate policies based upon how well they align with our preferences,

that is going to be operationalized by reducing a KL divergence between our preferences and observations.

And that task of optimization of the incremental reduction of divergence relative to our generative model of incoming observations,

that's what we want to solve, and it also turns out to have some good computational properties.

So, Jonathan?

Yeah, I was just going to say, just in terms of the equation that, sorry, excuse me, for the expected free energy,

although I feel very comfortable in general with mathematics and physics,

I find it very hard to get intuition from that equation, and I think it would be useful,

or maybe it's something that I'm going to do, but to actually run through a sort of really concrete example at that stage,

you know, with numbers, so you can see how the two terms differ for a particular example.

I think that would be quite enlightening for me to go through at some stage.

Thanks.

I know some may have notebooks that do that,

and we have, so however people want to share that or build that, that sounds good.

Yeah, many ways to go.

Like, but great questions.

Thank you for sharing these.

So, KL plays nicely with logs in the sense that it's defined essentially in terms of the divergence between Q to P.

Again, not the distance between Q and P.

The divergence between Q to P is defined as triangle equals expectation over the variational distribution on hidden states.

So, over the map, not over the territory.

Expectations over the map, so the center of gravity of our map's inference on hidden states,

of the difference between Q of X and P of X.

The log, again, connects us to all of those nice mathematical computational statistical properties of logs

and has this divergence interpretation too.

Expectation of a difference of logs is a KL divergence.

And the reason why it's not symmetric is because P to Q would be flipped,

and so it'd be a different number, unless they were literally both zero.
Okay.
Okay.
Okay.
Under some conditions, one can reduce variational free energy to other notions.
Which conditions?
Here they're showing it for expected free energy, these kinds of special reduction conditions
or special cases, and something analogous can be proposed for variational free energy.
It's just not directly shown there.
So in variational free energy, F, Q, and Y, expected free energy, epistemic value, pragmatic value.
And the special cases are when, for example, pragmatic value, which here is shown as the first term,
and then here, epistemic value is the second term.
So where section one is zero, this term is just zeroed out, there's no preferences.
Preferences are flat.
All outcomes are equally pragmatically valuable.
Then, we only have elements 2, 3, 4, 5, and play.
That is equivalent to Bayesian Surprise novelty search optimal information gain.
In contrast, if we had discarded 2, 3, 4, 5, because it's a fully discovered, fully known situation,
there's no epistemic value to harvest, then we only have the left term, one,
which is classical, pragmatic decision theory, and so on.
By acting on the world to change the way in which data are generated,
we can ensure a model is fit for purpose by choosing those data which are least surprising under our model.
Is active inference the process that can explain treatment resistance?
I can kind of imagine what that is, but if somebody who wrote that wants to describe what treatment resistance is.
Okay, under just this first pass understanding that treatment resistance,
let's just copy this information in,
has to do with recalcitrance or resilience to a given therapeutic intervention,
there might be different ways to frame it,
but if there's a model with a low learning rate,
I am the kind of person who does this action.
Then, from the clinician's perspective,
various interventions might be characterized,
the person may be characterized as resistant to those interventions
because the interventions are not updating some aspect of their generative model.
So, is it the process can explain treatment resistance?
Well, again, that's like asking, is linear regression
the process that can explain the relationship between heart disease and cancer?

It's like, I don't know,
if you write the paper and the community of experts use it,
then it's a going scientific explanation.
Is just putting these words down the explanation?
No.
What else needs to be built?
How can that question be modeled and engaged with?
But does the framework itself simply explain it?
No.
Could you explain it and attempt to build a model
that provides unique explanations, predictions, and control options?
I believe so.
Any comments or questions on this?
Okay.
Can someone read Bayes' rule for me once again
in plain English, equation 2.1,
now taking into consideration that, quote,
Bayesian inference minimizes surprisal,
surprise,
is equivalent to maximizing Bayesian model evidence?
And then they're suggesting that this is computing the posterior.
Any thought on this?
Okay.
equation 2.1.
Maybe if anyone wants to type it out in the coming weeks,
just, I'm not 100% sure,
but if you have the model that's parameterized tuned,
such that Bayesian model evidence is maximized,
you have found the most likely parameter,
the maximum likelihood estimate,
for a given sequences of actions given your priors.
So you have the most supported model,
so you have the most supported model.
That is equivalent to saying that with your most supported model,
you've minimized your surprise.
So,
so,
how you read different terms and like which one you see as kind of upstream or feeding into the other one

depends on the structure of what your experiment actually is.
page 24.
It then generates an action,
sends it to the environment in an attempt to make the environment less surprising.
Could this be translated to a clinical scenario as follows?
1.
I predict bodily integrity.
2.
I receive sensory observations disconfirming my prediction.
3.
I generate an action undergoing treatment to receive sensory observations
that confirm my prediction of bodily integrity,
thereby making my environment less surprising,
aka active inference.
Anyone have a thought on that?
For qualitative act-inf,
sounds good.
Let's just contrast that with other stories that people could tell.
One would be the reward learning story.
I'm rewarded by bodily integrity.
I seek out actions that increase my reward.
So,
I decided to go undergo treatment to increase my reward.
Act-inf,
generative model made a prediction.
Then,
based upon sensory observations,
policies were selected
to reduce surprise.
There's some interplay of change your mind and change the world.
The other story,
3,
this is 3A,
and then 3B,
is,
I got the sensory observation disconfirming my prediction.
I updated my prior.
I updated my prior.

Now,
expect low bodily integrity.
issue solved.
So,
normative in how we describe statistical problems,
sure.
Normative in terms of
in the human clinical environment,
suggesting how to act,
certainly not.
but,
great start.
policies associated with lower expected free energy,
G,
are assigned higher probability,
and become the policies that an organism expects to pursue.
Is this a reason why,
in a therapeutic setting,
change is often so hard to achieve?
Because the person often has to select different policies than they normally do,
and these new policies are therefore assigned lower probability.
Great question.
Any thought?
G,
updates,
the policy prior.
So,
we can look at,
we're going to come to this more in chapter 5,
when we start talking.
Thanks,
Jay.
We're going to talk about it more,
in chapter 5,
there was a physical human.
In case that didn't make any sense.
We're going to talk about it more,
in

5.
E
is the,
can be seen as the policy prior.
It's
the
list of affordances that are possible,
and the number in each of those cells,
is how likely those
are to occur,
a priori.
So,
I turn right 60% of the time,
I turn left 40% of the time.
A priori,
if nothing else intervenes,
I'm going to continue to replicate that 60-40 behavioral pattern.
expected free energy evaluation,

sharpens that
policy prior,
into a policy posterior,
that better reflects,
expected free energy minimization,
as we just discussed,
which can be decomposed,
into the epistemic and pragmatic value.
So,
various ways of thinking about it,
but
certainly,
in certain
settings,
if one has a strong habit,
they have a precise prior on action,
then,
yes,
it would be hard to update,

because they have a precise prior
on action.
But,
in the end,
it would come down to the specific model that you actually make
of that
therapeutic setting.
But again,
just qualitatively,
conversationally,
that broadly makes sense.
Especially if there's a new policy introduced,
one can imagine that it has a low prior,
because empirically,
it's never been seen.
Like,
how can one be the kind of person who runs every day,
if they haven't observed themselves doing that?
Any other comments on that?
Can I just ask you,
if there's a zero prior,
can it still be updated,
such that there is a non-zero posterior?
Yes.
That is going to come in,
with the Dirichlet conjugate prior,
we were just talking about it in the chapter seven,
and that's what's called learning by counting.
So,
in that setting,
you basically,
like,
each time you observe the two options,
you drop one ball of that color into an urn,
and then your posterior is just sampling from that urn.
So,
if you have one blue and one green,
then the third ball is going to,

like,

update your posterior a lot.

Whereas,

if you had a thousand of each,

then that two thousandth and first ball is not going to update it that much.

So,

that's,

like,

more recalcitrant to learning.

And then,

now someone says,

oh,

now there are orange balls.

Okay,

I observed an orange ball.

Okay,

now we have one orange ball,

and a thousand blue,

and a thousand green.

now we have one orange ball,

and a thousand blue,

and a thousand green.

now we have one orange ball,

one reason why those models are not,

like,

those models are very situational.

Like,

there's no general answer or solution about,

like,

adding.

And also,

from a computational perspective,

like,

that,

you need to start,

some of the variables need to change their dimensionality.

And so,

currently,

tools are being developed to facilitate,

specifying and updating the dimensionality of these models.

Because a lot of the things that people are just intuitively curious about are,

like,

well,

what happens when somebody gets a piece of information that helps them realize that a new affordance is possible,

and that new affordance then,

in the future,

may or may not open up another affordance.

It's like,

yeah,

cool situation to model.

Not going to have a general solution at this point.

And last of the pre-stated questions.

38,

they say,

VFE minimization is the key outside loop of ACT-INF,

sufficient to optimize perception and beliefs about policies.

However,

the notion of policy was introduced in the context of EFE,

which is an imagined future thing,

whereas VFE is right now.

How can beliefs about policies come into VFE without EFE?

Any thoughts on this?

Expected free energy is explicitly calculated over policies about future observations.

Variational free energy,

as the question suggested,

is about the current status of the variational distribution

and incoming data.

So they're just calculating something slightly different.

Without EFE,

we simply have energy-based learning via minimization of EFE.

Only by calculation of expected free energy over policy

do we get the active part of active inference.

Yes.

Expected free energy could be applied at the one time step horizon.
I'm sure there's other nuances and ways to address it.
But broadly,
variational free energy is reactive
in that it's an ongoing,
unfolding flow between the generative model and the incoming data.
But it can be calculated of active or passive entities.
Expected free energy makes a proactive...
Yeah,

and VFE,
it could describe
unplanned,
reflexive,
optimal activity.
Whereas to explicitly consider policy counterfactuals as such
and evaluate sequences of future possibilities,
only EFE has that capacity.
Okay.
Great.
Great question.
Scott,
when one models a candle flame or pendulum,
will they only use VFE?
Yeah,
so some years ago...
I'll let you go for it first.
Yeah,
just about VFE and EFE.
Well,
actually,
EFE
includes both planning,
I mean,
action selection planning,
and also decision making.

variational free energy is just about...

I mean,
it's not just about planning.
So
EFE also
includes
the action selection or policy selection.
And,
of course,
it's a requirement for planning and any kind of decision making.
But in VFE,
it's just
an inference we get
by
minimizing that
parameter.
but in the next stage,
when we want to act upon
that inference,
we use EFE for
the action selection
and decision making
and so on.
Thank you.
Yes.
A few answers on
using a candle flame.
So here,
in the strange particles paper,
thinking about a modeling iterative process,
first,
first,
we could say
we're going to model it
as an inert particle.
And then,
if there's no statistical variance
left to describe,
you've done

good enough.
But you might also need to appeal
to a latent cause.
Like,
oh,
it's like,
it's a guitar string
that's resonating,
but it's also like
there's like a
little gnome
with a circadian rhythm
latent cause
inside.
And then it's like
the next level
of strange particles,
like,
oh,
this guitar string
is resonating
and there's a circadian gnome
and I think the gnome
thinks about
what I'm going to do.
But each of those
are an empirical question
about what level
of sophistication
of cognitive model
is relevant
for a given situation.
And yes,
earlier versions,
before expected free energy
was even introduced,
ACDIMPH was
VFE based

and EFE was proposed
to explicitly account
for counterfactuals
and planning
because
in the earliest iterations
it was like,
well,
we'll just calculate VFE.
That's like the real
variable anyway.
So let's just do VFE
on generative models
and then like
just sort of be guided
step by step
and then EFE
like kind of
ultimately describes
and invokes
variables that VFE doesn't
like expected future
observations.
Great
questions.
Just want to
quickly
look through
where we're going to go
in chapter three.
We can,
of course,
come back to it
for the coming two weeks,
but any questions
or thoughts
or structuring
or notes

that people want to add
leading up to it,
of course,
the more,
the better.
So,
beginning with
a Dawkins quote
about survival
machines.
Chapter two
was
free energy
as a means,
free energy
as a how,
variational Bayesian inference
as a how.
here,
free energy
is going to be
approached
from the why,
which is going to be
the avoidable
of surprising states
and the ways
that that
is manifest
across like
self-organized systems
and self-modeling systems.
Mark off blankets
are introduced.
I'm sure we'll have
a lot of fun
discussions,
so please,

all questions
that people have
on the topic,
just write them out.
you leverage it
by like 100,000
when you actually
write it out
and improve
others' questions
because there are
many questions
that we have
about these areas.
Mark off blankets
are defined
a certain way
in a certain way
that's carrying
less philosophical
baggage
than many
initially suspect.
The action
perception loops,
which were
presented as loops
in the previous
chapter,
are here
presented
in what's known
as the
particular partition.
It's a particular
partition
because it's a joke.
First off,

it's a specific
way to do it.
It's not the only
way to do it.
It's particular.
Also,
what it partitions
is called
a particle.
The particle
is the blanket
and internal states.
That's why
these are called
inert,
active,
conservative,
and strange particles.
The particle
is the generative
model
of the agent.
So,
the particular
partition
helps us
operationalize
that action
perception
loop
scheme
into
a more
proper
Bayesian
graph
setting,
which enables

message passing,
all these other
things.
Figure 3.2
is
showing
a simple
example
of how
Markov blankets
are used
in inference,
not in some
super recursive
clinical setting,
just in terms
of like
some stochastic
variables
and the ways
that different
variables have
information about
one another,
and particularly
how internal
and external
states
can still
contain information
about each
other,
despite being
mediated only
through the
blanket.
3.3
agents with

Markov blankets
appear to model
the external
environment
in the sense
that internal
states correspond
to a
representation
of external
states of
the system.
So,
there's not
like a little
thermometer
inside of
the head
of the
generative
model,
but there
are some
statistical
patterns
such that
we can say
that there's
mutual
information
between
some part
of the
temperature
and some
part of
the generative
model.

Otherwise,
the agent's
actions
would be
insentient
with respect
to that
environmental
variable.
So,
that's like
where we
get
cybernetics,
good regulator
theorem,
law of
requisite
diversity.
There's a
turn towards
or a
nod to
some of
the
Bayesian
physics
by framing
the process
of surprise
minimization
as a
Hamiltonian
principle of
least action.
We're going
to talk
about it

more.
Short
summary,
least action
does not
mean most
lazy.
If you
shoot a
ball out
of a
cannon
and it
follows a
parabola,
that is
the path
of least
action.
So,
least action
does not
mean least
ATP
using,
it doesn't
mean least
carbohydrate
burning,
it doesn't
mean smallest
computational
model,
it's a
technical
physics
term that
has a

very important
meaning,
and you'll
hear Friston
and others
characterize
FEP as
a principle
of least
action for
cognitive
systems.
3.4
relationship
between
inference,
cognition,
and stochastic
dynamics.
Like,
why are we
even talking
about cognition
as inference
and talking
about all
of this
statistical
stochastic
stuff?
So,
here they
juxtapose
statistical
physics,
which has
long had
an imperative

of minimizing
free energy,
with
Bayesian
inference
and information
theory,
which can
be mobilized
to provide
cognitive
interpretations
under the
Bayesian
brain
framework.
Free
energy,
again,
we're not
simply talking
about ATP
being burnt.
It's not
Gibbs free
energy,
it's not
calories,
but it
might have
some
relationships.
3.41,
variational
free energy,
model evidence
and surprise,
fancy

letter,
surprise,
as model
evidence.
We'll come
back to it.
Expected
free energy,
here we
had VFE
and surprise,
real-time
unfolding
model evidence
as surprise.
Here's
prospective,
EFE,
an inference
to the
most likely
trajectory.
And
3.5,
active
inference,
a novel
foundation
to understand
behavior
and cognition.
Some
summary
of the
above
and a
little bit
of unpacking

about how
behavior
and cognition
are being
unified
in active
inference.
3.6,
models,
policies,
and trajectories.
More
discussion
about those
topics.
reconciliation
of inactive
cybernetic
and predictive
theories under
active
inference.
Those
three
apparently
disconnected
theoretical
perspectives
are
reconciled
in this
section,
according to
the authors.
And then
moving even
further,
active

inference
from the
emergence
of life
to agency
is where,
again,
in only
several
short
paragraphs,
so just
the tip
of the
iceberg,
there's
some
discussion
of
general
sentience
and
agentic
properties
of the
generative
models,
the kind
that we're
modeling
with active
inference.
Sidebox
on entropy
minimization?
Yes,
hardly any
maths in

chapter 3.
Yeah,
let's see.
Not in the
second half.
Okay.
Equation
3.2.
Equation
3.1,
but both of
them are
kind of
definitional
and surely
are described
naturally
in the text.
definition
in the
box 3.
So yes,
different style,
but it's a
breather before
chapter 4
hits hard
again with
the math.
So,
thank you
all.
Looking forward
to the
comments and
talk to you
next week.